

OMAR ALKHAMISSI

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EDUCATION

Fanshawe College, London, ON

Expected December 2026

Advanced Diploma, Computer Programming and Analysis

GPA: 3.91/4.2

- Awards: Dean's Honour Roll (two-time recipient)
- **Relevant Coursework:** Data Structures & Algorithms, OOP (Java/C++), Advanced Databases (MSSQL), Client/Server Application Development, Software Design Patterns, Full Stack Programming, Mobile Development

TECHNICAL SKILLS

Languages: C#, JavaScript, TypeScript, Python, SQL

Frontend: React 18/19, Vue 3 + Quasar, MUI, Vite, Axios, Socket.IO, React Router

Backend: ASP.NET Core, Node.js, Express, Entity Framework Core, MongoDB

Databases: MongoDB, Microsoft SQL Server, MySQL

Tools: Git, GitHub Actions, Azure, JWT, Stripe API, Jest, xUnit

PROJECTS

Debug My Heart | MERN Full-Stack Web App | *React, Node/Express, MongoDB, Stripe*

April 2026

- Built a **5,921-LOC MERN application** end-to-end — **React** frontend, **Express** REST API, and **MongoDB** schema — featuring a weighted **5-factor compatibility matching algorithm** returning normalized 0–100% scores with ranked result sets.
- Implemented **JWT authentication** with **bcryptjs** and **React Context** token persistence, securing **14 pages and 13 API routes** across **6 MongoDB models** with role-based middleware.
- Integrated **Stripe Subscription API** with idempotent checkout sessions, and built a **6-suite Jest + Supertest** harness using **mongodb-memory-server** for isolated integration tests.

FastFoodOrdering | Full-Stack Ordering Platform | *ASP.NET Core 8, EF Core 9, Vue 3 + Quasar*

August 2025

- Built an **ASP.NET Core 8 Web API** with **EF Core 9 Code-First migrations** across 5 relational tables, exposed via **Swagger** and consumed by a **Vue 3 + Quasar** frontend.
- Implemented **RFC2898DeriveBytes** password hashing (10,000 iterations, 64-byte salt) and **JWT HMAC-SHA256** authentication, configurable via appsettings.json.
- Designed an **async DB transaction** (BeginTransactionAsync → CommitAsync) atomically creating Tray records, inserting line items, and computing 9 nutrition aggregate fields.

Real-Time Chat | Multi-Room Messaging App | *Express, Socket.IO 4, React 19, MUI 7*

April 2026

- Built a **real-time multi-room chat** application with **Socket.IO 4 + Express 5** supporting per-room user lists, typing indicators, and soft-delete edits via /edit and /del commands.
- Designed a **Room class** with a private static **#rooms Map** enabling unlimited concurrent rooms on demand, plus a 90-color user palette with deduplication and release on disconnect.
- Built a **React 19 + MUI 7** client with **useRef** socket persistence, day-divider auto-injection via **date-fns**, and dual-thread design separating user input from background server events.

EXPERIENCE

Freelance Front-End Developer, Remote

April 2024 – December 2025

- Delivered responsive, **SEO-friendly** websites in HTML, CSS, and JavaScript for **3 small-business clients**, shipping multi-page sites tested across **mobile, tablet, and desktop breakpoints**.
- Implemented **WCAG accessibility standards** and cross-browser compatibility fixes, improving Lighthouse accessibility scores into the **90+ range** on shipped pages.
- Managed client feedback cycles and shipped **incremental updates within 48-hour turnarounds** based on revision requests.

LEADERSHIP & ACTIVITIES

Fanshawe College AI Club, Engagement Member

November 2024 – Present

- Contribute to hands-on machine learning workshops covering practical model building and collaborate with peers on AI/ML projects exploring classification and NLP techniques.
- Help coordinate technical workshops and guest speaker sessions for a community of 50+ members.

Liberty International High School Student Council, President, Alexandria, Egypt

January 2021 – October 2022

- Led a 12-member council and introduced a structured planning process that grew event participation and improved peer feedback collection.
- Represented student concerns to administration, advocating for scheduling and policy changes adopted school-wide.